

ORIGINAL ARTICLE

The safety and feasibility of probiotics in children and adolescents undergoing hematopoietic cell transplantation

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Hematopoietic cell transplantation (HCT) has become a standard treatment for many adult and pediatric conditions. Emerging evidence suggests that perturbations in the microbiota diversity increase recipients' susceptibilities to gut-mediated conditions such as diarrhea, infection and acute GvHD. Probiotics preserve the microbiota and may minimize the risk of developing a gut-mediated condition; however, their safety has not been evaluated in the setting of HCT. We evaluated the safety and feasibility of the probiotic, *Lactobacillus plantarum* (LBP), in children and adolescents undergoing allogeneic HCT. Participants received once-daily supplementation with LBP beginning on day –8 or –7 and continued until day +14. Outcomes were compliance with daily administration and incidence of LBP bacteremia. Administration of LBP was feasible with 97% (30/31, 95% confidence interval (CI) 83–100%) of children receiving at least 50% of the probiotic dose (median 97%; range 50–100%). We did not observe any case of LBP bacteremia (0% (0/30) with 95% CI 0–12%). There were not any unexpected adverse events related to LBP. Our study provides preliminary evidence that administration of LBP is safe and feasible in children and adolescents undergoing HCT. Future steps include the conduct of an approved randomized, controlled trial through Children's Oncology Group.

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INTRODUCTION

Hematopoietic cell transplantation (HCT) has become a standard treatment for many adult and pediatric malignant and nonmalignant conditions, with over 60 000 patients undergoing HCT each year.¹ Treatment with HCT is associated with a variety of morbidities such as non cancerous organ or tissue dysfunction, decreased quality of life and physical function, secondary cancers, increased susceptibility to infections and acute or chronic GvHD.^{1,2} Therefore, innovative supportive care interventions that reduce the toxicity associated with HCT are essential to reduce treatment-related morbidities and further improve survival.

The intestinal microbiota critically functions in maintaining the physical, functional and immunologic barriers within the gastrointestinal (GI) tract such that perturbations in the diversity of the microbiota result in aberrant systemic immune responses as well as pathogen colonization and mucosal invasion.³ A significant relationship has been observed between insults to the GI tract and increased incidence of bacteremia and GvHD among HCT recipients.^{4–8} These data suggest that not only translocation into the host tissues is important in the development of systemic infections, but also the interaction between the intestinal microbiota and the host immune mechanisms are important in the development of other serious complications.⁸ Furthermore, loss of microbial diversity has recently been shown to be an independent risk factor for mortality following allogeneic HCT.⁶ These investigations suggest that interventions that preserve the microbiome may prevent HCT morbidities and improve survival.

Probiotics are nutritional supplements that contain a defined amount of viable microorganisms and upon administration confer a benefit to the host.⁹ Clinical trials and systematic analysis have suggested that probiotics are safe and may be beneficial for immunocompromised adults, adults with cancer receiving abdominal radiation and a wide range of pediatric conditions.^{10–17} Although some studies have not demonstrated a benefit of probiotic therapy, systematic reviews and meta-analysis have attributed much of the inconsistency in the literature to the testing of variable doses, duration and strains.^{15,18,19} Moreover, there has been a lack of studies elucidating the mechanisms of probiotics in a controlled clinical setting. There remains a need for studies on probiotics that advance both their clinical application and biological effects in the host.

Lactobacillus plantarum (LBP) is one of the most thoroughly studied species and has been investigated in a variety of clinical settings, including immunocompromised adults and children, and has a wide safety profile.^{20–27} Supportive evidence exists that replenishing commensals such as *Lactobacillus* before murine HCT substantially decreases acute GvHD severity and intestinal insult as well as prevents the sequelae of a number of other HCT morbidities.^{28,29} Reintroduction of *Lactobacillus* during the transplant period has also been associated with decreased GvHD lethality and pathology, an effect that may be related to *Lactobacillus* colonization preventing the expansion of *Enterococcus* in the gut.⁷

Probiotics may have a benefit in the clinical setting; however, there have been concerns about the safety of probiotics in

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severely immunocompromised patients, particularly those undergoing HCT. To address this concern and before the conduct of a randomized, controlled, clinical trial, we performed a pilot study to determine the safety and feasibility of the probiotic, *LBP*, in children and adolescents undergoing myeloablative allogeneic HCT.

SUBJECTS AND METHODS

Study design and objective

A convenience sample of 30 children and adolescents undergoing HCT during the period of January 2010 to June 2014 was recruited to the study. The primary objective of this study was to determine the safety of administering the probiotic, *LBP*, to children and adolescents undergoing HCT. The study was approved by the institutional review board of each participating institution and was conducted in accordance with the provisions of the Declaration of Helsinki. Parents or legal guardian provided informed consent for each participant. For children aged ≥ 7 years, informed assent was obtained. The trial was registered at Clinicaltrials.gov (NCT01010867) and was conducted and monitored by the Federal Drug Administration (FDA) as an investigational new drug (108977). The study was open for a period of 4 years, a time frame that was longer than anticipated. This was because of the process of obtaining an investigational new drug, ensuring representation of each age group (mandated by the FDA) and significant personnel changes that occurred during the midpoint of the study. These factors resulted in minimal patient recruitment for > 1 year.

Patient selection

Children and adolescents between the ages of 2 and 17 years undergoing first myeloablative allogeneic HCT were eligible for participation. A requirement by the FDA was that one-third of the patients were between 2 and 3.99 years of age. Eligible donor sources were HLA-matched sibling or parent, a related donor mismatched for a single HLA locus (class I or II), unrelated bone marrow or PBSC donor or unrelated cord blood at least 4/6 antigen match (class I or II). The sources of stem cells were bone marrow, umbilical cord blood or cytokine-mobilized peripheral blood. Exclusion criteria consisted of: (1) self-prescribed probiotic therapy within 3 months of admission for the conditioning regimen; (2) allergy to oats; (3) a diagnosis of inflammatory bowel syndrome, short small bowel syndrome, Crohn's disease or ulcerative colitis; (4) patients with a history of bowel resection; and (5) prophylactic broad-spectrum antibiotics. An amendment to permit the concomitant use of levofloxacin and probiotics was approved on 23 April 2014 to avoid barriers to accrual to the pilot study. Sensitivity analysis was performed to ensure *LBP* was not sensitive to levofloxacin (data not shown). Participants received the institutional standard-of-care for GvHD prophylaxis.

L. plantarum

Participants received *LBP* (Probi, Lund, Sweden) orally or by an enteral feeding tube at a dose of 1×10^8 colony-forming units/kg/day. All shipments were accompanied by a Certificate of Analysis that documented the viability of bacteria and that the product was free of bacterial and fungal contamination. An antibiotic susceptibility profile was performed on each shipment of *LBP* before patient entry onto study and bacterial viability testing was performed to ensure stability of the colony-forming units. *LBP* was dispensed by the research pharmacy and began on day -8 or -7 , depending on the conditioning regimen, and continued through day $+14$, a time frame coinciding with insult to the intestinal mucosa and concomitant neutropenia. Compliance to the prescribed regimen was measured by examination of the medication administration record or other hospital source documentation.

Clinical assessments and safety monitoring

Participant diagnosis and conditioning regimen was abstracted from the medical record. From the initiation of the conditioning regimen until day $+28$, we recorded the incidence of positive blood cultures and documented infections; Grade 3–5 Common Terminology Criteria for Adverse Events (version 3.0) unexpected nonhematologic toxicities; incidence and etiology of diarrhea; incidence and grade of acute GvHD; incidence of *LBP* bacteremia; and the incidence of *Clostridium difficile*

(diagnosed by toxin B PCR). Weekly blood cultures were required to screen for *LBP* on day -6 , day 0, day $+7$ and day $+15$. A select group of participants (22 of the 30) were screened for colonization of *LBP* in the stool on a weekly basis using Remel MRS Agar (Lenexa, KS, USA). In the case that a patient became febrile or developed persistent diarrhea, additional blood or stool cultures were obtained as per institutional guidelines. The incidence and grade of acute GvHD and survival was monitored until day $+100$.

Statistical methods

The primary safety end point was the incidence of *LBP* bacteremia. Patients were assessed for this end point from the day treatment started (initiation of therapy) until day $+28$. The study design would accept at most one *LBP* bacteremia among 30 patients. A secondary end point was the feasibility of administration of *LBP*. Feasibility was defined as receiving at least 50% of the probiotic dose, with the design target of $\geq 75\%$ of the study participants receiving at least 50% of the probiotic dose. The incidence of non-*Lactobacillus* bacteremia through day $+28$, new onset of *C. difficile* through day $+28$ and the incidence of grade 3 or 4 acute GvHD through day $+100$ were also assessed as secondary end points. All enrolled patients were included in the assessment of the feasibility end point. For the primary safety end point and the secondary clinical end points, an evaluable patient needed to receive at least 50% of the dose of probiotics. Descriptive statistics were calculated for patient demographics (mean/SD and median/range for age, count/proportion for other categorical variables). Proportional incidences of the outcome variables were calculated along with the 95% exact binomial confidence intervals (CIs).

RESULTS

Thirty-one patients were entered into the trial. The demographics of the 30 evaluable study participants are presented in Table 1. Slightly more males than females were recruited to the study. Racial background was Caucasian (53%), African American (40%) and Asian (7%). Nearly half of the transplants were performed for the treatment of a malignant disease (43%), the remaining transplants were due to a range of hematologic disorders.

Safety and feasibility of *LBP*

Administration of *LBP* was feasible with 97% of enrolled children (30/31, 95% CI 83–100%) receiving at least 50% of the probiotic dose. Among the 30 evaluable patients, the mean number of doses taken was 92% (SD 14%) and the median was 97% (range 50–100%) (Table 2). No episodes of *LBP* bacteremia (0% (0/30; 95% CI 0–12%)) were observed. The incidence of non-*Lactobacillus* bacteremia was 20%. New onset of *C. difficile* was observed in 20% of participants. We did not observe any serious adverse events or unexpected severe adverse events attributed to *LBP* in any patient enrolled to the study. One patient with severe aplastic anemia developed acute appendicitis on day $+28$. Pathologic examination of the appendix revealed a predominance of Gram-negative enteric organisms and only a small number of Gram-positive organisms that were morphologically inconsistent with *Lactobacillus*. This patient tolerated the appendectomy and after discussion with the pathologist, it was determined that the appendicitis was unlikely due to *Lactobacillus*. The patient fully recovered and was discharged from the hospital.

Three patients died before day $+100$ and none of these deaths were attributed to *LBP*. The causes of death included veno-occlusive disease, idiopathic pneumonia syndrome and multi-organ system failure in one patient, disease progression in the second patient and complications due to aspiration pneumonia in the third patient.

Acute GvHD

The majority of participants ($N=21$; 70%) did not develop acute GvHD by day $+100$. None of the participants who died before day $+100$ developed acute GvHD. Of the 9 patients diagnosed with

Table 1. Demographic and clinical characteristics

Characteristic	Evaluable patients (N = 30)
Institution	
All Children's Hospital, Johns Hopkins Medicine	12 (40%)
New York-Presbyterian Morgan Stanley Children's Hospital, Columbia University Medical Center	17 (57%)
Nemours Children's Hospital	1 (3%)
Age at enrollment (years)	
Mean (SD)	7.7 (4.7)
Median (range)	6.9 (2.2–17.3)
Age at enrollment (years)	
2–3.99	10 (33%)
≥ 4	20 (67%)
Gender	
Female	14 (47%)
Male	16 (53%)
Race	
African American	12 (40%)
Asian	2 (7%)
White	16 (53%)
Diagnosis	
Sickle cell	9 (30%)
Malignancy–leukemia	12 (40%)
Malignancy–lymphoma	1 (3%)
Severe aplastic anemia	4 (13%)
Thalassemia	2 (7%)
Fanconi anemia	1 (3%)
Myeloproliferative disorder	1 (3%)
Donor	
Mismatched unrelated cord	2 (7%)
Mismatched unrelated donor	5 (17%)
Matched related donor	12 (40%)
Matched umbilical cord	2 (7%)
Matched unrelated donor	9 (30%)
Stem cell source	
Cord blood	5 (17%)
Marrow	22 (73%)
PBSCs	3 (10%)
Preparatory regimen	
Busulfan–fludarabine-based regimens	13 (43%)
Busulfan–melphalan-based regimens	3 (10%)
Fludarabine–melphalan-based regimens	3 (10%)
TBI–TLI-based regimens	5 (17%)
Cyclophosphamide based regimens	6 (20%)

acute GvHD before day +100, 5 patients (17%) developed grade 2 and 4 (13%) developed grade 3. No patients were diagnosed with grades 1 or 4 acute GvHD. Acute GI GvHD was observed in 23% of the patients ($N=7$). Three (10%) children were classified with stage 1 and 4 children (13%) with stage 3. No patient developed stage 4 acute GI GvHD.

Colonization of LBP

To screen for colonization of LBP in the stool, up to four weekly stool samples were obtained in 22 of 30 participants and tested for the presence of LBP. The majority had at least one stool specimen that was positive for *Lactobacillus* (21/22, 95%). Despite this restricted analysis, 77 specimens (75%) were positive

Table 2. Clinical outcomes

Clinical variable	Evaluable patients (N = 30)
Compliance (% dose received)	
Mean (SD)	92% (14%)
Median (range)	97% (50–100%)
Serious adverse event	
Yes	0 (0%)
No	30 (100%)
<i>Lactobacillus</i> bacteremia by day 28	
Yes	0 (0%)
No	30 (100%)
Non-<i>Lactobacillus</i> bacteremia by day 28	
Yes ^a	6 (20%)
<i>Staphylococcus epidermidis</i>	1
<i>Serratia marcescens</i>	1
<i>Enterococcus faecium</i>	1
<i>Klebsiella pneumoniae</i>	3
<i>Streptococcus viridans</i>	1
<i>Fusarium species</i>	1
No	24 (80%)
New-onset <i>Clostridium difficile</i> by day 28	
Yes	6 (20%)
No	24 (80%)
Stage of gastrointestinal (GI) GvHD by day 100^b	
0	23 (77%)
1	3 (10%)
2	0 (0%)
3	4 (13%)
4	0 (0%)
Site and stage of non-GI GvHD by day 100^b	
Skin	
Stage 1	0 (0%)
Stage 2	2 (6%)
Stage 3	2 (6%)
Stage 4	0 (0%)
Liver	
Stage 1	0 (0%)
Stage 2	1 (3%)
Stage 3	0 (0%)
Stage 4	0 (0%)
Overall grade of GvHD by day 100^b	
0	21 (70%)
1	0 (0%)
2	5 (17%)
3	4 (13%)
4	0 (0%)

^aTwo patients developed two non-*Lactobacillus* infections. ^bThree patients died before day 100 without developing GvHD.

for *Lactobacillus*, suggesting GI colonization with administration of LBP.

DISCUSSION

To our knowledge, this is the first clinical trial using the probiotic, LBP, during the conditioning regimen and the post-HCT neutropenic period in children and adolescents undergoing myeloablative HCT. Although probiotic usage has been shown to be safe in a variety of other clinical settings,^{12,13,16,26,27} several case reports suggest that the probiotic itself may have caused bacteremia in the setting of HCT.^{30–32} Our results suggest that

LBP administration is both safe and feasible. In contrast to the published case reports, we are assured that the probiotic used for this study was free from contamination and maintained viable colony-forming units throughout the study period. However, we do not recommend routine probiotic administration based on the results of this study. Our next step is to conduct a multicenter, double-blind, randomized, controlled trial that has received approval by the Children's Oncology Group. This study will examine the impact of probiotic therapy on clinical outcomes and composition of the microbiome.

This pilot study provides preliminary support for the safety probiotic administration beginning on the first day of conditioning therapy and ending on day +14. No instances of *LBP*-bacteremia or *LBP*-related severe adverse events were reported. Of the 30 evaluable patients, 3 died before day +100; however, none of these deaths were related to the administration of *LBP*. We found that ~20% of children developed non-*Lactobacillus* bacteremia between day 0 and day +30, a figure similar to the published literature.^{33–35} Our study did report one case of appendicitis. Despite consensus from the data safety monitoring board, study investigators and pathologist that this was unlikely related to the administration of *LBP*, it cannot be ruled out. This potential risk factor will be monitored in our subsequent clinical trial.

Our study also found that administration of probiotics was well tolerated with a compliance rate that was much higher than other pediatric supportive care studies in HCT.³⁶ Excellent compliance may be related to a once-daily administration, a small amount of liquid and a highly motivated patient population as many patients undergoing HCT self-prescribe probiotic therapy. These results suggest that if probiotics are found to be effective and safe, patients may be largely compliant to probiotic therapy because of the perceived benefits of probiotics and its extensive safety profile.

Patients undergoing allogeneic HCT receive immunosuppressive medications, antibiotics and other interventions that can cause diarrhea. We found that 20% of our population developed *C. difficile*. The average incidence of new-onset *C. difficile* in pediatric HCT has not been defined in a large trial. Among adult HCT recipients, the incidence of *C. difficile* has been reported to be between 4 and 20%, a figure in line with our observed incidence.³⁷

An exploratory aim of this study was to evaluate the association of probiotic therapy with the incidence of acute GvHD because of recent reports describing the association of perturbations of the microbiome and the onset of acute GvHD.^{6,8,29} We found that 70% of evaluable patients did not develop acute GvHD, a number that is comparable to reported rates in other allogeneic trials.³⁸ However, our study has many important distinguishing features. Our study was performed in a universal pediatric population with all patients <18 years of age and exposed to myeloablative conditioning regimens. Sixty percent of children received cells from an unrelated donor that has been identified as a risk factor for acute GvHD when compared with trials using only matched sibling donors. Our study provides reasonable evidence that the use of probiotics did not result in an increase in the incidence of acute GvHD.³⁹ However, the lack of a comparable group within the published literature combined with our small sample size prevents a conclusive statement to be determined. We hypothesize that the longer use of the probiotics post HCT may result in nearly universal gut colonization that could decrease the development of acute GvHD. Future randomized trials will confirm or refute these hypotheses.

The findings of this pilot study are limited by several factors. The study of a single probiotic rather than a multistrain probiotic was predicated on the preclinical and clinical evidence available at the time of study development. *LBP* was identified as the ideal probiotic because of its safety record among immunocompromised patients, evidence supporting biological plausibility and preclinical evidence supporting the beneficial role of *Lactobacillus*

in the setting of HCT. Multistrain probiotics may impart a more beneficial effect on the microbiome and may be associated with different clinical outcomes; however, there remains controversy surrounding the beneficial use of single compared with multistrain products.¹⁹ Additional trials are warranted. Our study was limited by our evaluation of *LBP* in the stool. Although this study provides reasonable assurance that *LBP* survived the GI tract, it does not prove colonization or provide information on the effects of *LBP* on the composition of the microbiome. This limitation has been recognized in the upcoming Children's Oncology Group trial that will include analysis of the microbiota at study entry and throughout the study period.

In the past 20 years, there has been little reduction in the acute GvHD rate in allogeneic transplantation. This is despite the introduction of many new agents administered in the prophylactic setting. Current research manipulating donor cells as well as novel cellular therapies may reduce the acute GvHD rate for some populations but realistically will not eliminate it entirely. Our data, combined with the existing literature, suggest that probiotics may play a role in reducing acute GvHD, transplant-associated non-*Lactobacillus* bacteremia, *C. difficile* and overall length of stay. However, these end points cannot be confirmed outside of the context of a larger clinical trial. Our pilot trial provides reasonable assurance to proceed with novel studies that are designed to further confirm the safety of probiotic therapy in the setting of HCT and to examine the efficacy on clinically meaningful outcomes.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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